ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** B gets hotter than A	Thin wire has higher resistance.
3	**(a)** the iron rod	Iron concentrates the field.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Series connection allows it to sense all current.
5	**(c)** glows dimmer	Reversed cell opposes the others.
6	**(b)** the current is larger	More current.
7	**(c)** A is true but R is false.	A is true; R is false because a thick wire has less resistance.
8	**(a)** strength rises with turns	Doubling turns gives about double pins.
9	**(d)** retains magnetism	Steel is used for permanent magnets.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Oersted's finding.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Both feet are at the same potential (1); No current passes through the bird (1).	1+1
12	Total resistance increases (1); Less current for each (1).	1+1
13	Current decreases (1); The electromagnet becomes weaker (1).	1+1
14	10 A fuse (1); It is just above 9 A, so it protects without blowing in normal use (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Increase the number of turns (1); Increase the number of cells (1); Use a thicker soft iron core (1).	1+1+1
16	Wire resistance is high (1); Less current reaches the bulb (1); Use a short thick wire (1).	1+1+1
17	The switch breaks the circuit (1); No current flows (1); But switching off the plug is also advised (1).	1+1+1
	OR: The heater (1); It carries a larger current (1); A thick wire has low resistance and does not overheat (1).	1+1+1
18	Both give deflection since current flows (1); Iron foil may change the field slightly (1); Paper has no effect (1).	1+1+1
	OR: Most energy goes to light, not heat (1); Filament is small (1); Not efficient (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Independent: number of turns; dependent: pins lifted (1); Constant: cell, nail, wire (1); Procedure: 10, 20, 30, 40 turns and count pins (1); Expected: pins rise with turns (1); Conclusion (1).	1+1+1+1+1
	OR: Cell about 1.5 V (1); Battery: several cells, higher voltage (1); Mains about 220 V (1); Low voltage cannot drive a harmful current through the body (1); Mains can cause fatal shock (1).	1+1+1+1+1
20	Total = 4.5 + 0.5 + 0.3 (1); = 5.3 A (1); Exceeds 5 A, so unsuitable (1); Fuse would melt in use (1); Choose a 10 A fuse for safety (1).	1+1+1+1+1
	OR: Risk of shock (1); Switch off at the main (1); Remove the plug (1); Use insulated tools (1); Test before touching (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** melt	Total 7 A exceeds 5 A.
2	**(c)** doubled	Heat is proportional to time.
3	**(c)** A is true but R is false.	A is true; R is false: it has a very high melting point.
4	**(c)** deflects from its north-south position	Current produces a magnetic field that acts on the needle.
5	**(a)** an electric shock	Live wire exposed.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Reason explains it.
7	**(b)** no current passes through the bird since there is no potential difference across its feet	No complete path.
8	**(d)** find and remove the cause such as overloading	Safety.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(d)** 9 A	Parallel branches add: 2+3+4 = 9 A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No, only a coil produces a strong field (1); The nail needs a coil of wire carrying current (1).	1+1
12	A long wire fits in less space as a coil (1); Compact design gives high resistance and heat in a small area (1).	1+1
13	It melts first on high current (1); Breaks the circuit before wires or appliances are damaged (1).	1+1
14	Energy is wasted as heat (1); Use thick wire of low resistance or high voltage transmission (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Live meets neutral so resistance falls (1); Very large current heats wires (1); Fuse melts; MCB trips (1).	1+1+1
16	Wire has resistance so heat is produced (1); Any moving charge makes a magnetic field (1); Both effects occur together (1).	1+1+1
17	Thin wire has high resistance (1); More heat is made per second (1); Thick wires of low resistance are used for connection (1).	1+1+1
	OR: Old cell supplies less current (1); Weaker magnetic field (1); Strength depends on current (1).	1+1+1
18	Count pins lifted (1); Change one factor at a time (1); Plot a graph (1).	1+1+1
	OR: Coil: carries current (1); Core: gets magnetised (1); Armature, hammer and gong: strike and sound (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Magnetic effect moves the armature (1); Heating from coil resistance (1); Sparks at contact (1); Wear of contact (1); Insulated coil limits heating (1).	1+1+1+1+1
	OR: Compass deflected by current (1); Link between electricity and magnetism (1); Electromagnets for cranes (1); Bells and relays (1); Foundation for motors and generators (1).	1+1+1+1+1
20	Resistance of wire converts electric energy to heat (1); Thin wire has more resistance (1); Overheating damages insulation (1); Use thicker wire for heavy loads (1); Fuses limit current (1).	1+1+1+1+1
	OR: In series same current (1); A has higher resistance, hotter (1); In parallel B draws a larger current (1); B still stays cooler due to low resistance (1); Heat depends on resistance and current (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** heating and magnetic	Kettle heats; bell uses magnetism.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(c)** continues to heat	Independent paths.
4	**(c)** so leakage current flows to the ground and not through the user	Safety measure.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Excess current causes fire.
6	**(d)** 15 A	Just above the load.
7	**(a)** in circles around the wire	Circular.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(b)** high safety against shocks and heat	Insulation.
10	**(c)** using a wire of lower resistance	Lower resistance gives less heat.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A coin does not melt at a low current (1); Appliance may overheat and cause a fire (1).	1+1
12	Soft iron demagnetises quickly (1); The hammer can reset quickly (1).	1+1
13	The third pin is connected to earth (1); It carries leakage current safely away (1).	1+1
14	(a) Heating effect (1); (b) and (c) Magnetic effect (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It connects earth first when plugging (1); Disconnects last on removal (1); Ensures safety (1).	1+1+1
16	Soft iron loses magnetism at once (1); Steel stays magnetised (1); Release must be quick (1).	1+1+1
17	Magnetic only when current flows (1); Strength changes with current and turns (1); Poles can be reversed (1).	1+1+1
	OR: To cut off the live supply (1); A fuse on neutral leaves the appliance live (1); Hence a danger of shock (1).	1+1+1
18	Both are electric currents (1); Cell gives a small voltage (about 1.5 V) (1); Mains is about 220 V and dangerous (1).	1+1+1
	OR: Use correct rating (1); Switch off before replacing (1); Never use wire or coin (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Test circuit with a gap (1); Conductor: bulb glows (1); Insulator: bulb stays off (1); Wires: copper inside, plastic outside (1); Tool handles: rubber/plastic (1).	1+1+1+1+1
	OR: Cell: chemical to electrical (1); Bulb: electrical to light and heat (1); Motor: electrical to motion (1); Chemicals are consumed (1); Cell gets used up (1).	1+1+1+1+1
20	More turns (1); More current (1); Bigger soft iron core (1); If current fails the load falls (1); Safety zone and backup systems (1).	1+1+1+1+1
	OR: Overloading (1); Short circuit (1); Faulty wiring (1); Use proper rated fuses/MCB (1); Regular inspection and no sharing of sockets (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.